

PROJECT REPORT ON

Advanced Folder Encryption

Submitted by

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Under the Supervision of

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COMPANY INTRODUCTION:

Supraja Technologies is a leading Knowledge and Technical Solutions Provider and pioneer leader in IT industry, is operating based out of Vijayawada.

R&D at Supraja

With a 24X7 work in Research & Development, experts at Supraja Technologies work under our :

- **Supraja Technologies Cyber Security Cell**

About Supraja Technologies:

Supraja Technologies (a unit of CHSMRLSS Technologies Pvt. Ltd.) with its foundation pillars as Innovation, Information and Intelligence is exploring indefinitely as a **Technology Service Provider (Corporate Consulting)** and as a **Training Organization (Ed-Tech)** as well.

You may visit us at :

www.suprajatechnologies.com

The multi domains of trainings which Supraja Technologies operate include the following :

• **Workshops & Hackathons**

- Engineering Colleges
- Corporate Companies (Startups & MNC's)
- Government Organizations

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An MSME & ISO 9001:2015 Certified Company

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Branch Office : Prabhu Villas, Plot No. 124, 1st Floor, Church Road, Ayyappa Nagar, Vijayawada - 520007.

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- **Classroom Trainings Cum Certification Courses**

- Summer Training (30-45 Days)
- Winter Training (10 - 15 Days)
- Weekend Training (2 Days)
- 1 Month / 3 Months / 6 Months Courses

- **On-site Trainings**

- Value Added Courses / Two Credit Courses for Colleges
- Faculty Development Programs
- College Summer Training (15 Days, 30 Days, 45 Days & 60 Days)
- Govt Agencies, Police Academies, Corporates etc

- **Cloud Campus**

- (Distance Learning Program) *Coming Soon

- **Internships**

- Internship for Engineering Students (30 Days, 45 Days & 60 Days)
- Internship for Graduates (6 Months)

- **Lab Setup**

- Cyber Lab

- **CoE**

- Cyber Security Centre of Excellence

- **Supraja Technologies Security Assessment Product**

- (Our SaS Product is currently under development) *Coming Soon

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Why Supraja Technologies:

Be it Training or a workshop, the course content is always from R&D Cell of Supraja.

- A proven track record of delivering quality services.
- **68,500+** Students trained by our trainers till date.
- Training Partners of recognized institutions.
- Trainers with excellent research and teaching pedagogy illustrate their findings through corporate standard **practical demonstrations** during their sessions.
- Easy to learn and **hands-on sessions** are given, with additional benefits of Study Material, Tool kit and immediate query handling.
- Self-Prepared **Cyber Security Cell**.
- Supraja Technologies has the best, experienced and highly **skilled bunch of R&D Engineers, Security Analysts, Security Consultants & Trainers**.
- We provide training in Innovating and Trending Technologies to Govt. Officials, Corporate Houses and Colleges.

✓ Something we are proud of :

1. Supraja Technologies CEO Mr.Santosh Chaluvadi is an Alumni of Potti Sriramulu Chalavadi Mallikharjuna Rao College of Engineering and Technology, Vijayawada.

With our CEO this college conducted/organised a 50 hours Nonstop Marathon Training Workshop on Ethical Hacking & Cyber Security for which this respective college and our CEO both holds their name in **"LIMCA BOOK OF RECORDS 2017"**

2. We are very happy to inform you all that our company, Supraja Technologies has been shortlisted for **"Top 50 Tech Companies" award 2019**, conferred at InterCon - Dubai, UAE.

Supraja Technologies is one out of thousands companies that were initially screened by InterCon team of 45+ research analysts over a period of three months and the final shortlist includes 150+ firms and we are very proud to inform you all that our company Supraja Technologies also happens to be a part of the same.

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✓ Life changing solution/service :

After working on R&D for around 2 years, finally in the mid 2019 we have successfully developed a service/solution of various techniques and strategies for the Film Industry through which he can kill piracy of any film in online up to 35% right now. This betaservice is being appreciated & adopted by various Tollywood Film Industry Producers & Hero's to safeguard their film from piracy in online and to gain more profits.

By the end of 2033 our vision is to rollout a complete full packed service/solution where we can kill piracy entirely 100% everywhere in online for sure.

Appreciation :

Received a great appreciation from our 1st Tollywood Film Industry client Mr.Saptagiri for providing our Anti-Piracy betaservice for his film VAJRA KAVACHADARA GOVINDA

✓ Achievements:

- On 17th August 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security (CoE)** at Ramco Institute of Technology, Rajapalayam
- On 18th September 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security (CoE)** at SRM University (Ramapuram Campus), Chennai
- On 20th November 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security (CoE)** at St.Joseph's Institute of Technology, Chennai

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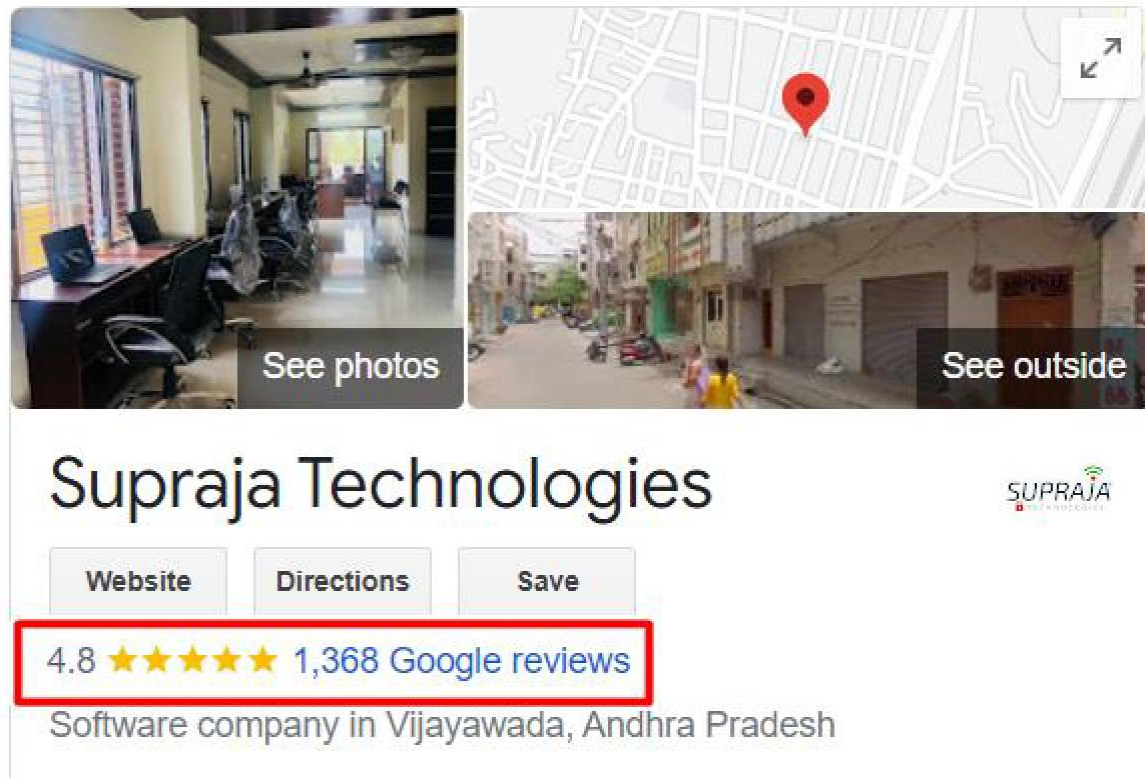
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**Our Company Supraja Technologies
is one of the emerging startup
in Andhra Pradesh
with **4.8** Google Ratings**



Link : <https://bit.ly/SuprajaGoogle>

AND

**also check our Company CEO Instagram Profile
for our recent more success stories:**

<https://www.instagram.com/chaluvadisantosh/>

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Santosh Chaluvadi

Founder & CEO
Supraja Technologies

He is a 28-year-old entrepreneur, one of the India's efficient Cyber Security Analyst and also he is an expert Digital Marketer as well. He is a digital marketer by profession and security enthusiast by passion. He primarily focuses on content building, testing and monetization of blogs. He has successfully developed many websites and done the security testing himself to ensure that the user's data is in safe hands and their privacy is protected. He is very active on social media and shares lot of tech stuff with his followers. The young student hacker has solved many issues with the vulnerabilities present in various websites and databases, given a solution in clearing the loopholes in order to protect the data to be leaked from the databases. Besides Ethical Hacking & Cyber Security, he also has a passion in Blogging & Digital Marketing.

While pursuing his engineering itself, he has trained many young generation people/students of more than 3500+ from various parts across Andhra Pradesh through his workshops, seminars, courses in Cyber Security and this makes him one of the youngest student trainers in India.

At the age of 20 he conducted his first workshop on Blogging & Ethical Hacking which was the beginning to his success in this field and right now he has handful of workshops to train students, government and corporate organizations as well in Andhra Pradesh & Telangana. He is the only student trainer who started conducting workshop for his peers, professors and for corporates.

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The year 2016 gave me the conviction and confidence to notch up whatever I was doing. I'd organized a 50-hour marathon event on Ethical Hacking and Cyber Security in PSCMR College, Vijayawada that went on to bag to achieve Limca Book of Records for non-stop longest duration workshop. The impact we were making was clearly visible by now. Diverse people from Jammu & Kashmir in the North to Kanyakumari in the South had attended the event. Some of the Government of India officials took notice of this record and invited our team for couple of conferences at New Delhi. The country's esteemed institutions were reaching out to me to conduct training, workshops, and events on a myriad of subjects related to online security. Multi-National Corporations (MNC's) had begun to consult me on the Cyber Security of their systems.

❖ Life changing solution/service :

After working on R&D for around 2 years, finally in the mid 2019 we have successfully developed a service/solution of various techniques and strategies for the Film Industry through which he can kill piracy of any film in online up to 35% right now. This betaservice is being appreciated & adopted by various Tollywood Film Industry Producers & Hero's to safeguard their film from piracy in online and to gain more profits.

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❖ Records, Appreciations, Awards & Recognitions etc at a glance :

1. Holds a National Record in “Limca Book of Records – 2017”
2. Ex-Associate Member for National Cyber Safety and Security Standards (NCSSS)
3. Steering Committee Member for United Conference on Cyber Space (UNITEDCON 2020)
4. Judge for the Grand Finale of SIH (Smart India Hackathon 2024) Software Edition for the Nodal Centre at Sri Sai Ram Engineering College, Chennai which is an initiative by Ministry of Education (Govt. of India) and AICTE
5. Received Appreciation from Mr.Amit Narayan, Executive Director at Rajahmundry Asset of Oil and Natural Gas Corporation Limited (ONGC) on 16th December, 2022 for training their employees on Cyber Security
6. Awarded as a "Karmaveer Chakra - 2019", on 12th October 2019 at IIT Delhi, which was instituted by iCONGO in partnership with the United Nations
7. Received Appreciation from Mr.Sandeep Rathore, Commissioner of Police, Chennai on 2nd March 2024 for the exceptional commitment and invaluable contribution as a Member of JURY at Greater Chennai Police Cyber Hackathon 3.0
8. Evaluator for the “Innovative Bharat” which is organized by AICTE and Ministry of Education held on 6th January 2024
9. Judge for the Grand Finale of SIH (Smart India Hackathon 2023) Senior Software Edition for the Nodal Centre at PVPSIT, Vijayawada which is an initiative by Ministry of Education (Govt. of India) and AICTE
10. Appointed as a Member for Board of Studies on 19th April 2025 for the Departments of Cyber Security, Data Science and AIML at Bapatla Engineering College, Bapatla
11. Appointed as a Member for Board of Studies on 12th March 2025 for the Department of Cyber Security at St. Joseph’s Institute of Technology, Chennai
12. Appointed as one of the Industry Academia Advisory Council Member on 30th September 2023 for the Department of Cyber Security at VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad
13. Appointed as a Member for Board of Studies on 4th May 2023 for the Department of Cyber Security at Madanapalle Institute of Technology & Sciences, Madanapalle
14. Appointed as a Member for Board of Studies for the Department of MCA at NRI Institute of Technology, Perecherla (Guntur)
15. Awarded as a “Social Media Influencer - 2019”, on 30th June 2019 by Jignasa in association with Government of Andhra Pradesh
16. Nominated for “INDIA 500 CEO AWARD 2019”
17. Invited & Interviewed by ETV Andhra Pradesh news channel on 27th July, 2019 for a Special Story Interview on “Spy Apps”
18. Appreciated by Mr.Sridhar, Sub-Inspector of Police at Central Crime Branch, Vijayawada on 23rd October, 2018 for exclusively training him on Special Investigation Course, which will help him to solve the cyber crime cases easily
19. Received a great appreciation from our 1st Tollywood Film Industry client Mr.Saptagiri, for providing Anti-Piracy betaservice for his movie VAJRA KAVACHADARA GOVINDA

❖ Something we are proud of :

We are very happy to inform you all that our company, Supraja Technologies has been shortlisted for "**Top 50 Tech Companies**" award 2019, conferred at InterCon - Dubai, UAE.

Supraja Technologies is one out of thousands of startup companies that were initially screened by InterCon team of 45+ research analysts over a period of three months and the final shortlist includes 150+ firms and we are very proud to inform you all that our company Supraja Technologies also happens to be a part of the same.

Some Glimpses of our Journey



Mr.Santosh Chaluvadi – CEO, Supraja Technologies
Giving hands-on Cyber Security training workshop to the CSE students
at IIT Kharagpur



Mr.Santosh Chaluvadi – CEO, Supraja Technologies was Invited & Interviewed by ETV Andhra Pradesh news channel on 27th July, 2019 for a Special Story Interview on “Spy Apps”



Mr.Santosh Chaluvadi – CEO, Supraja Technologies was awarded as a "Social Media Influencer 2019" in recognition of his remarkable achievements in the social media as a part of First International Social Media Festival on 30th June 2019 by Jignasa in association with Government of Andhra Pradesh

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On 23rd October 2018 Mr.Santosh Chaluvadi, CEO - Supraja Technologies and Mr.Krishna Chaitanya, CTO - Supraja Technologies had successfully completed delivering Special Investigation Course training in Cyber Security to Mr.Sridhar Garu, Sub-Inspector of Police at Central Crime Branch, Vijayawada which will help him to solve the cyber crime cases easily



ETV Andhra Pradesh News Channel interviewed Mr.Santosh Chaluvadi, CEO - Supraja Technologies for his achievements in the domain of Cyber Security & Digital Marketing



Supraja Technologies was invited by Indian Air Force (Air Wing NCC) to deliver a session on Latest Cyber Crimes & Awareness for the NCC cadets, staff and officers on 4th July, 2019



Supraja Technologies – CEO, CTO & CMO with Indian Air Force (Air Wing NCC) Group Captain Sandeep Gupta.
We thank Mr.Sandeep Gupta for inviting us on 4th July, 2019 to deliver a session on Latest Cyber Crimes & Awareness for the Indian Air Force (Air Wing NCC) cadets, staff & officers

ADVANCED FOLDER ENCRYPTION SYSTEM

Project Report

Submitted by

Sravani Poola

B.Tech, Computer Science Engineering

Madanapalle Institute of Technology & Science

In association with Supraja Technologies

Emp ID: ST#IS#9790

Team No: 3

ADVANCED FOLDER ENCRYPTION SYSTEM

Project Report

Submitted by

Anusha Guthikonda

B.Tech, Computer Science Engineering

Madanapalle Institute of Technology & Science

In association with Supraja Technologies

Emp ID: ST#IS#9789

Team No: 3

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1. Abstract

The Advanced Folder Encryption System is a Python application that encrypts and decrypts files using the AES algorithm. It provides a Tkinter graphical user interface to select folders, enter a password, and perform secure encryption and decryption.

2. Introduction

Data security is important for protecting sensitive information. This project uses AES encryption to keep files secure and prevent unauthorized access.

3. Objectives

- Secure folders using AES.
- Provide a simple Tkinter GUI.
- Enable password-based encryption and decryption.
- Keep the operation easy and secure.

4. Software Requirements

- Python 3.x
- Tkinter
- PyCryptodome
- Windows 10 / 11

5. Hardware Requirements

- Intel / AMD Processor
- 4 GB RAM
- 100 MB Free Storage

6. Modules

File	Description
main.py	Tkinter GUI — the application entry point
encrypt.py	Handles folder encryption logic
decrypt.py	Handles folder decryption logic
aes_security.py	Core AES encryption / decryption engine
requirements.txt	Lists required Python libraries
run.bat	Batch script to run the project

7. Working

1. Run main.py.
2. Browse and select the target folder.
3. Enter the password.
4. Click “Encrypt Folder” to secure its contents.
5. Click “Decrypt Folder” to restore the original files.

8. Advantages

- Strong AES-based encryption.
- Easy-to-use graphical interface.
- Fast processing.
- Password-protected access.

9. Limitations

- No password recovery mechanism.
- Works on the local system only.

10. Future Enhancements

- OTP-based authentication.
- Cloud backup support.
- User login system.
- Dark mode interface.

11. Source Code

The complete implementation is organized into four modules: the Tkinter GUI, the encryption handler, the decryption handler, and the shared AES engine.

main.py

```
import tkinter as tk
from tkinter import filedialog, messagebox
from encrypt import encrypt_folder
from decrypt import decrypt_folder

def browse_folder():
    folder = filedialog.askdirectory()
    folder_var.set(folder)

def encrypt_action():
    folder = folder_var.get()
    password = password_entry.get()
    if folder == "":
        messagebox.showerror("Error", "Please Select Folder")
        return
    if password == "":
        messagebox.showerror("Error", "Please Enter Password")
        return
    encrypt_folder(folder, password)

def decrypt_action():
    folder = folder_var.get()
    password = password_entry.get()
    if folder == "":
        messagebox.showerror("Error", "Please Select Folder")
        return
    if password == "":
        messagebox.showerror("Error", "Please Enter Password")
        return
    decrypt_folder(folder, password)

root = tk.Tk()
root.title("Advanced Folder Encryption System")
root.geometry("550x450")
root.configure(bg="white")
root.resizable(False, False)

folder_var = tk.StringVar()

title = tk.Label(
    root,
    text="Advanced Folder Encryption System",
    font=("Arial", 18, "bold"),
    bg="white",
    fg="blue"
)
title.pack(pady=20)

tk.Label(
    root,
    text="Folder Path",
    bg="white",
```

```

        font=("Arial", 11)
    ).pack()

    frame = tk.Frame(root, bg="white")
    frame.pack(pady=5)

    folder_entry = tk.Entry(
        frame,
        textvariable=folder_var,
        width=40
    )
    folder_entry.pack(side=tk.LEFT, padx=5)

    browse_button = tk.Button(
        frame,
        text="Browse",
        command=browse_folder
    )
    browse_button.pack(side=tk.LEFT)

    tk.Label(
        root,
        text="Password",
        bg="white",
        font=("Arial", 11)
    ).pack(pady=15)

    password_entry = tk.Entry(
        root,
        show="*",
        width=35
    )
    password_entry.pack()

    encrypt_btn = tk.Button(
        root,
        text="Encrypt Folder",
        command=encrypt_action,
        bg="green",
        fg="white",
        width=20
    )
    encrypt_btn.pack(pady=15)

    decrypt_btn = tk.Button(
        root,
        text="Decrypt Folder",
        command=decrypt_action,
        bg="blue",
        fg="white",
        width=20
    )
    decrypt_btn.pack()

    exit_btn = tk.Button(
        root,
        text="Exit",
        command=root.destroy,
        bg="red",
        fg="white",
        width=20
    )
    exit_btn.pack(pady=20)

    root.mainloop()

```

encrypt.py

```
import os
from tkinter import messagebox
from aes_security import AESCipher

def encrypt_folder(folder_path, password):
    if not os.path.exists(folder_path):
        messagebox.showerror(
            "Error",
            "Selected folder does not exist!"
        )
        return

    cipher = AESCipher(password)
    encrypted_count = 0

    try:
        for root, dirs, files in os.walk(folder_path):
            for file in files:
                file_path = os.path.join(root, file)
                try:
                    with open(file_path, "rb") as f:
                        data = f.read()
                    encrypted_data = cipher.encrypt(data)
                    with open(file_path, "wb") as f:
                        f.write(encrypted_data)
                    encrypted_count += 1
                except Exception:
                    pass

        messagebox.showinfo(
            "Success",
            f"{encrypted_count} file(s) encrypted successfully."
        )
    except Exception as e:
        messagebox.showerror(
            "Encryption Error",
            str(e)
        )
```

decrypt.py

```
import os
from tkinter import messagebox
from aes_security import AESCipher

def decrypt_folder(folder_path, password):
    if not os.path.exists(folder_path):
        messagebox.showerror(
            "Error",
            "Selected folder does not exist!"
        )
        return

    cipher = AESCipher(password)
    decrypted_count = 0

    try:
        for root, dirs, files in os.walk(folder_path):
            for file in files:
                file_path = os.path.join(root, file)
                try:
```

```

        with open(file_path, "rb") as f:
            encrypted_data = f.read()
            decrypted_data = cipher.decrypt(encrypted_data)
            with open(file_path, "wb") as f:
                f.write(decrypted_data)
            decrypted_count += 1
    except Exception:
        pass

    messagebox.showinfo(
        "Success",
        f"{decrypted_count} file(s) decrypted successfully."
    )
except Exception as e:
    messagebox.showerror(
        "Decryption Error",
        str(e)
    )

```

aes_security.py

```

import base64
import hashlib
from Crypto.Cipher import AES
from Crypto.Random import get_random_bytes
from Crypto.Util.Padding import pad, unpad

class AESCipher:
    def __init__(self, password):
        self.key = hashlib.sha256(password.encode()).digest()

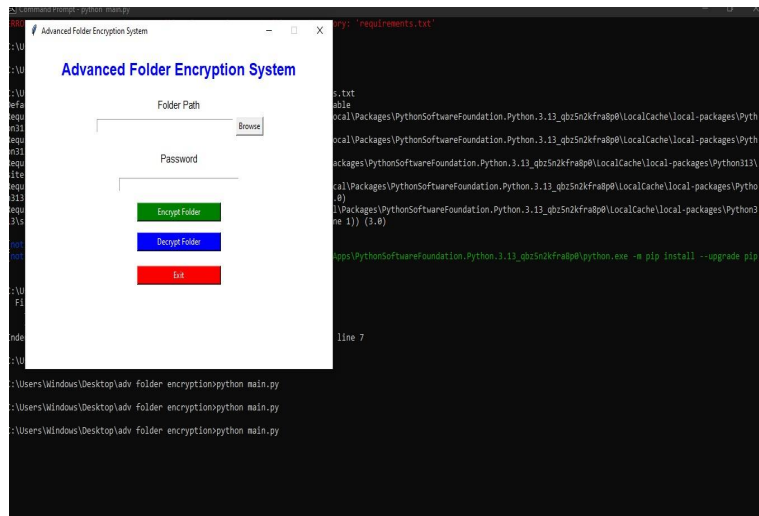
    def encrypt(self, data):
        iv = get_random_bytes(16)
        cipher = AES.new(self.key, AES.MODE_CBC, iv)
        encrypted_data = cipher.encrypt(pad(data, AES.block_size))
        return base64.b64encode(iv + encrypted_data)

    def decrypt(self, encrypted_data):
        encrypted_data = base64.b64decode(encrypted_data)
        iv = encrypted_data[:16]
        ciphertext = encrypted_data[16:]
        cipher = AES.new(self.key, AES.MODE_CBC, iv)
        decrypted_data = unpad(
            cipher.decrypt(ciphertext),
            AES.block_size
        )
        return decrypted_data

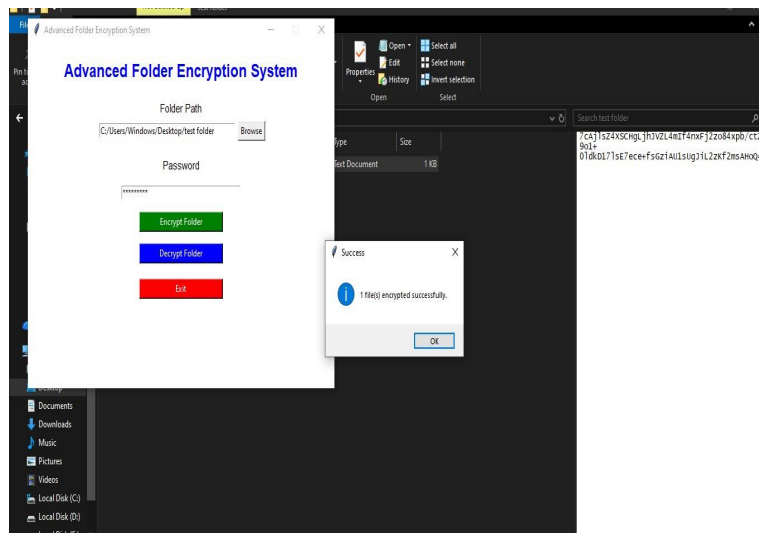
```

12. Output Screenshots

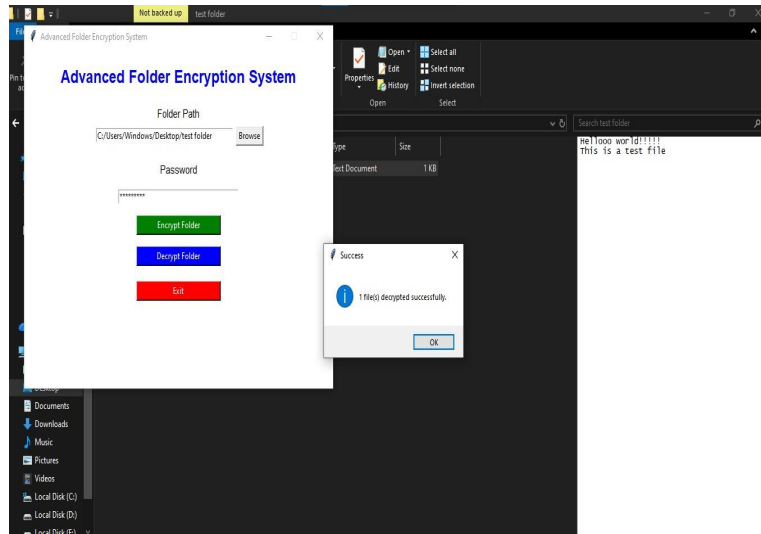
Application window before an operation is run:



Successful folder encryption:



Successful folder decryption, with the original file contents restored:



13. Conclusion

The project successfully secures folders using AES encryption with a user-friendly Tkinter interface, providing a straightforward way to protect confidential files through password-based encryption and decryption.

14. References

- Python Documentation — docs.python.org
- Tkinter Documentation — docs.python.org/3/library/tkinter.html
- PyCryptodome Documentation — pycryptodome.readthedocs.io